

Chapter 6

Least square approximation:

A well defined system has equal number of variables & equations

Example:

$$x_1 + 2x_2 + x_3 = 0$$

$$x_1 - 9x_2 + 7x_3 = 2$$

$$2x_1 + 3x_2 + 5x_3 = 5$$

$$\begin{matrix} A & x & b \\ \begin{bmatrix} 1 & 2 & 3 \\ 1 & -9 & 7 \\ 2 & 3 & 5 \end{bmatrix} & \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} & = \begin{bmatrix} 0 \\ 2 \\ 5 \end{bmatrix} \end{matrix}$$

$n \times n$ $n \times 1 \rightarrow n \times 1$

If we have a system where
no. of equations > no. of variables $\rightarrow$ overdetermined system

Example:

$$x_1 + 2x_2 + x_3 = 0$$

$$x_1 - 9x_2 + 7x_3 = 2$$

$$x_1 + 3x_2 + 5x_3 = 4$$

$$2x_1 + 11x_2 - 9x_3 = 5$$

$$x_1 + x_2 - x_3 = 7$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & -9 & 7 \\ 1 & 3 & 5 \\ 2 & 11 & -9 \\ 9 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 4 \\ 5 \\ 7 \end{bmatrix}$$

$m \times n$ $n \times 1$ $m \times 1$

Least square approximation ~~is a~~ method is a way to find an approximate solⁿ of an over-determined system.

$$S = (s_{ij})^T \quad O = (o_{ij})^T \quad C = (c_{ij})^T$$

$$A \cdot x = b$$

$\hat{C}$ = column to ori

$$O = (o_{ij})^T = (o_{ij})^T + (o_{ij})^T = (o_{ij})^T$$

How to solve?

$$S = (s_{ij})^T \quad \hat{C} = (s_{ij})^T \quad s_{ij} \times 10 + 10 = (s_{ij})^T$$

$$A^T \cdot A \cdot x = A^T \cdot b$$

[A^T multiplied on both sides]

$$(n \times m) \cdot (m \times n) \cdot (n \times 1) \quad (n \times m) \cdot (m \times 1)$$

square matrix
so can be inverted

$$\downarrow$$

$$n \times n$$

$$\downarrow$$

$$n \times 1$$

$$\downarrow$$

$$n \times 1$$

Example: $a_0 + a_1 x$ [no. of co-efficient = 2]

We want to fit a st. line through the following nodes

$$\begin{array}{lll} x_0 = -3 & x_1 = 0 & x_2 = 6 \\ f(x_0) = 0 & f(x_1) = 0 & f(x_2) = 2 \end{array}$$

no. of nodes = 3

$$P_1(x_0) = a_0 + a_1(x_0) = f(x_0) \Rightarrow a_0 + 3a_1 = 0$$

$$P_2(x_1) = a_0 + a_1(x_1) = f(x_1) \Rightarrow a_0 = 0$$

$$P_3(x_2) = a_0 + a_1(x_2) = f(x_2) \Rightarrow a_0 + 6a_1 = 2$$

$$\begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$\downarrow$ $\downarrow$ $\downarrow$
 1×1 1×2 2×2

Multiplying A^T on both sides

$$\begin{bmatrix} 1 & 1 & 1 \\ -3 & 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -3 & 0 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 3 \\ 3 & 45 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

Now you can apply gaussian elimination / LU / inverse method to find values of a_0 and a_1 .

inverse method:

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 45 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$= \begin{bmatrix} 3/7 \\ 5/21 \end{bmatrix}$$

$$P_1(x) = 3/7 + 5/21 x$$

Vector dot product:

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matrix notation:

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad y = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$x^T \cdot y = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$= 1 \times 4 + 2 \times 5 + 3 \times 6$$

$\downarrow$
= returns a scalar/number

vector notation:

$$\vec{x} \cdot \vec{y}$$

$$= (1 \times 4) + (2 \times 5) + (3 \times 6)$$

$\downarrow$

= returns a scalar/number

Inner product on l_2 -norm :

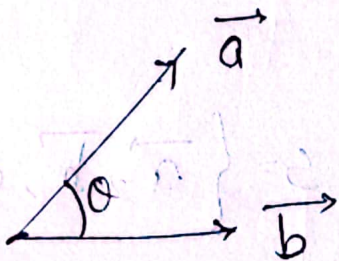
$$\vec{x} \cdot \vec{x} \text{ or } \vec{x}^T \vec{x}$$

Magnitude of a vector:

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$|a| = \sqrt{(a_1)^2 + (a_2)^2 + (a_3)^2}$$

Dot product (second approach):



$$a \cdot b = |a||b| \cos \theta$$

$$\cos \theta = \frac{a \cdot b}{|a||b|}$$

$$0 = \sqrt{2} \cdot \sqrt{2} |a||b|$$

$$0 = \sqrt{2} \cdot \sqrt{2}$$

$$0 = \sqrt{2} \cdot \sqrt{2}$$

Orthogonality: -

If $\angle$ between 2 vectors $= 90^\circ$,

or if 2 vectors are perpendicular to each other
they are orthogonal.

if they are orthogonal then their dot product
will be 0.

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta \quad [\cos 90^\circ = 0]$$

$$\therefore \vec{a} \cdot \vec{b} = 0$$

$$\therefore \vec{a}^T \cdot \vec{b} = 0$$

Let S is set of vectors $S = \{ \vec{a}, \vec{b}, \vec{c} \}$

Now S is orthogonal set iff

$$|\vec{a}| |\vec{b}| \vec{a} \cdot \vec{b} = 0$$

$$\vec{b} \cdot \vec{c} = 0$$

$$\vec{c} \cdot \vec{a} = 0$$

so each vector is perpendicular to each other

Orthonormality:

For being orthonormal they have to:

1) be orthogonal (dot product = 0)

2) length = 1 (Unit vectors)

$$a = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

$$\vec{a} \cdot \vec{b} = \vec{a} \cdot b = 4 \times 1 + 2 \times -3 + 1 \times 2 = 0$$

$\therefore$ orthogonal

$$|a| \neq 1 \text{ and } |b| \neq 1$$

So normalizing: $\hat{a} = \frac{\vec{a}}{|a|} = \frac{1}{\sqrt{21}} \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4/\sqrt{21} \\ 2/\sqrt{21} \\ 1/\sqrt{21} \end{bmatrix}$

$$|\hat{a}| = \sqrt{\left(\frac{4}{\sqrt{21}}\right)^2 + \left(\frac{2}{\sqrt{21}}\right)^2 + \left(\frac{1}{\sqrt{21}}\right)^2} = 1$$

$$\hat{b} = \frac{\vec{b}}{|\vec{b}|} = \frac{1}{\sqrt{14}} \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{14} \\ -3/\sqrt{14} \\ 2/\sqrt{14} \end{bmatrix}$$

$$|\hat{b}| = \sqrt{\left(\frac{1}{\sqrt{14}}\right)^2 + \left(-\frac{3}{\sqrt{14}}\right)^2 + \left(\frac{2}{\sqrt{14}}\right)^2} = 1$$

the process of converting vectors to unit vectors is called normalization.

So normalization change magnitude but not direction.

So now since they are orthogonal and has magnitude = 1, so they are orthonormal.

Example:

$$S = \left\{ \frac{1}{\sqrt{5}} (2, 1)^T, \frac{1}{\sqrt{5}} (1, 2)^T \right\}$$

$$S = \left\{ \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$\downarrow$
 u_1

$\downarrow$
 u_2

$$\vec{u}_1 \cdot \vec{u}_2 = \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}} + \frac{1}{\sqrt{5}} \times -\frac{2}{\sqrt{5}} = 0$$

[$\therefore$ orthogonal]

$$|\vec{u}_1| = \sqrt{\left(\frac{2}{\sqrt{5}}\right)^2 + \left(\frac{1}{\sqrt{5}}\right)^2} = 1$$

$$|\vec{u}_2| = \sqrt{\left(\frac{1}{\sqrt{5}}\right)^2 + \left(-\frac{2}{\sqrt{5}}\right)^2} = 1$$

[$\therefore$ Orthonormal].

$$Q^{-1} = Q^T$$

$$Q \cdot Q^T = I \quad \left\{ \frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right\} = 2$$

$$Q^T \cdot Q = I$$

if Q has column columns that are from orthonormal sets.

$$0 = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \sqrt{2} \cdot \sqrt{2}$$

[Laplace's]

$$1 = \left(\frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{2}} \right) = \sqrt{2}$$

$$1 = \left(\frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{2}} \right) = \sqrt{2}$$

[Laplace's]

QR decomposition:

Gram-Schmidt Process:

lets have a basis $(u_1, u_2, \dots, u_n)$ form a vector space. Gram-Schmidt process takes the basis $(u_1, u_2, \dots, u_n)$ and forms new orthogonal basis $(p_1, p_2, \dots, p_n)$. We can later transform these orthogonal basis into orthonormal basis $(q_1, q_2, \dots, q_n)$.

original basis: u_1, u_2, u_3
↓ Gram Schmidt Process

orthogonal basis: p_1, p_2, p_3

↓ normalization
Orthonormal basis: q_1, q_2, q_3

$$1) P_1 = U_1$$

$$2) P_2 = U_2 - \frac{U_2 \cdot P_1}{P_1 \cdot P_1} P_1$$

$$3) P_3 = U_3 - \frac{U_3 \cdot P_1}{P_1 \cdot P_1} P_1 - \frac{U_3 \cdot P_2}{P_2 \cdot P_2} P_2$$

Example:

$$U_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, U_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, U_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$1) P_1 = U_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$2) P_2 = U_2 - \frac{U_2 \cdot P_1}{P_1 \cdot P_1} P_1$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{(1 \times 1) + (0 \times -1) + (1 \times 1)}{(1 \times 1) + (-1 \times -1) + (1 \times 1)} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{2}{3} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 2/3 \\ -2/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix}$$

$$3) P_3 = U_3 - \frac{U_3 \cdot P_1}{P_1 \cdot P_1} P_1 - \frac{U_3 \cdot P_2}{P_2 \cdot P_2} P_2$$

$$= \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{(1 \times 1) + (1 \times -1) + (2 \times 1)}{(1 \times 1) + (-1 \times -1) + (1 \times 1)} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$- \frac{(1 \times 1/3) + (1 \times 2/3) + (2 \times 1/3)}{(\frac{1}{3} \times \frac{1}{3}) + (\frac{2}{3} \times \frac{2}{3}) + (\frac{1}{3} \times \frac{1}{3})} \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{2}{3} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} - \frac{5}{2} \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2/3 \\ -2/3 \\ 2/3 \end{bmatrix} - \begin{bmatrix} 5/6 \\ 5/3 \\ 5/6 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \end{bmatrix}$$

P_1, P_2, P_3 are orthogonal basis.

Now for normalization:

$$1) q_1 = \frac{P_1}{|P_1|} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$2) q_2 = \frac{P_2}{|P_2|} = \frac{1}{\sqrt{6/3}} \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} \sqrt{6}/6 \\ \sqrt{6}/3 \\ \sqrt{6}/6 \end{bmatrix}$$

$$3) q_3 = \frac{P_3}{|P_3|} = \frac{1}{2\sqrt{2}} \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -\sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix}$$

$$\begin{bmatrix} e^{i\phi} \\ e^{i\phi} \\ e^{i\phi} \end{bmatrix} \cdot \frac{(e^{i\phi} \times e^{i\phi}) + (e^{i\phi} \times e^{i\phi}) + (e^{i\phi} \times e^{i\phi})}{(e^{i\phi} \times e^{i\phi}) + (e^{i\phi} \times e^{i\phi}) + (e^{i\phi} \times e^{i\phi})} =$$

$$\begin{bmatrix} e^{i\phi} \\ e^{i\phi} \\ e^{i\phi} \end{bmatrix} \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$\begin{bmatrix} e^{i\phi} \\ e^{i\phi} \\ e^{i\phi} \end{bmatrix} \cdot \frac{1}{\sqrt{6/3}} \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} \sqrt{6}/6 \\ \sqrt{6}/3 \\ \sqrt{6}/6 \end{bmatrix}$$

QR decomposition:

Any $m \times n$ matrix A can be written in the form:

$$A = QR$$

where Q is a $(m \times n)$ matrix with orthonormal columns
 R is an upper triangular matrix of shape $(n \times m)$

$$A_{m \times n} = Q_{m \times n} \times R_{n \times m}$$

$$Q^T A = Q^T Q R$$

$$Q^T A = I R \quad [A A^T = I]$$

$$R = Q^T A$$

Example:

$$x_1 = -3$$

$$x_2 = 0$$

$$x_3 = 6$$

$$f(x_1) = 0$$

$$f(x_2) = 0$$

$$f(x_3) = 2$$

$$\begin{bmatrix} 1 & 0 & 6 \\ 1 & 0 & 6 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\downarrow \quad \downarrow$$

$$P_1 = U_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$P_2 = U_2 - \frac{U_2 \cdot P_1}{P_1 \cdot P_1} P_1 = \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix} + \frac{(-3 \times 1) + (0 \times 1) + (6 \times 1)}{1 \times 1 + 1 \times 1 + 1 \times 1} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -4 \\ -1 \\ 5 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad P_2 = \begin{bmatrix} -4 \\ -1 \\ 5 \end{bmatrix}$$

$$q_1 = \frac{P_1}{|P_1|} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$q_2 = \frac{P_2}{|P_2|} = \frac{1}{\sqrt{42}} \begin{bmatrix} -4 \\ -1 \\ 5 \end{bmatrix} = \begin{bmatrix} -4/\sqrt{42} \\ -1/\sqrt{42} \\ 5/\sqrt{42} \end{bmatrix}$$

$$Q = \begin{bmatrix} 1/\sqrt{3} & -4/\sqrt{42} \\ 1/\sqrt{3} & -1/\sqrt{42} \\ 1/\sqrt{3} & 5/\sqrt{42} \end{bmatrix}$$

$$R = Q^T A = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix}$$

$$Ax = b$$

$$A^T A \cdot x = A^T \cdot b$$

$$(QR)^T \cdot (QR)x = (QR)^T b$$

$$R^T Q^T \cdot Q R x = R^T Q^T b$$

$$R^T \cdot I \cdot R x = R^T Q^T b$$

$$R x = Q^T b$$

$$R x = Q^T b$$

$$\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & 42 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & 42 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 2\sqrt{3} \\ 5/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 3/7 \\ 5/21 \end{bmatrix}$$